MATHS MANTHAN

• TUITION CLASSES •

e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 15 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M, 3M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If three numbers in AP have sum 9 and product 24, then the numbers are: [1]

(A) $2, 3, 4$ (B) $1, 3, 5$ (C) $2, 4, 6$ (D) $3, 4, 5$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 2, 3, 4** [0.5 Mark]

Let the three terms be $(a-d), a, (a+d)$:

Formula / Theorem Used: $(a-d)+a+(a+d)=3a$ and $a(a^2-d^2)=24$

Justification: [0.5 Mark]

$3a=9 \implies a=3$, and $3(9-d^2)=24 \implies 9-d^2=8 \implies d^2=1 \implies d=\pm 1$. Numbers: $\mathbf{2, 3, 4}$.

Q2. The 4th term from the end of the AP $-5, -2, 1, \ldots, 49$ is: [1]

(A) 37 (B) 40 (C) 43 (D) 46

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 40** [0.5 Mark]

Given: $d=-2-(-5)=3$ and $l=49$.

Formula / Theorem Used: n^{th} term from end $= l-(n-1)d$

Justification: [0.5 Mark]

4th term from end $= 49-(4-1)(3)=49-9=\mathbf{40}$.

Q3. If $a_n = 5n-3$, then the sum of the first 20 terms (S_{20}) is: [1]

(A) 980 (B) 990 (C) 1000 (D) 1020

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 990** [0.5 Mark]

Given: $a_1=5(1)-3=2$ and $a_{20}=5(20)-3=97$.

Formula / Theorem Used: $S_n=\frac{n}{2}(a_1+a_n)$

Justification: [0.5 Mark]

$S_{20}=\frac{20}{2}(2+97)=10\times 99=\mathbf{990}$.

Q4. If a, b, c are in AP $(d \neq 0)$, then $\dfrac{a-b}{b-c}$ equals: [1]

(A) 1 (B) $\dfrac{a}{c}$ (C) $\dfrac{b}{a}$ (D) $\dfrac{c}{a}$

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 1** [0.5 Mark]

Since a, b, c form a non-constant AP with common difference d:

Formula / Theorem Used: $b - a = c - b = d \implies a - b = b - c = -d$

Justification: [0.5 Mark]

$\frac{a-b}{b-c} = \frac{-d}{-d} = \mathbf{1}$.

Q5. **Assertion (A):** The sequence $0.2, 0.22, 0.222, 0.2222, \ldots$ does not form an AP.
Reason (R): Because $a_2 - a_1 = 0.02 \neq 0.002 = a_3 - a_2$, the consecutive differences are unequal. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true, and (R) explains (A).** [0.5 Mark]

Testing the constant-difference definition:

Formula / Theorem Used: $a_2 - a_1 \neq a_3 - a_2 \implies$ Not an AP

Justification: [0.5 Mark]

$0.22 - 0.20 = 0.02$, whereas $0.222 - 0.220 = 0.002 \neq 0.02$. Both statements are true and (R) directly proves (A).

Q6. Find the sum of all odd integers between 2 and 100 that are divisible by 3. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying the AP and Number of Terms (n) [1 Mark]

Odd multiples of 3 between 2 and 100: $3, 9, 15, \ldots, 99$ ($a = 3$, $d = 6$, $l = 99$).

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + l)$

$3 + (n-1)(6) = 99 \implies 6(n-1) = 96 \implies n - 1 = 16 \implies n = 17.$

Step 2: Computing S_{17} [1 Mark]

$$S_{17} = \frac{17}{2}(3 + 99) = \frac{17}{2} \times 102 = 17 \times 51 = \boxed{\mathbf{867}}$$

Q7. Find the first negative term of the AP $53, 48, 43, \ldots$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Writing Inequality $a_n < 0$ [1 Mark]

Here $a = 53$ and $d = 48 - 53 = -5$. For the first negative term:

Formula / Theorem Used: $a_n < 0 \iff a + (n-1)d < 0$

$$53 + (n-1)(-5) < 0 \implies 58 - 5n < 0 \implies 5n > 58 \implies n > 11.6$$

Step 2: Concluding Term Index and Value [1 Mark]

Least integer $n > 11.6$ is $n = 12$. (Term $a_{12} = 53 - 55 = -2$.)

$$\boxed{\mathbf{12}^{\text{th}} \text{ term } (\text{value } = \mathbf{-2})}$$

Q8. Divide 56 into four parts in AP such that the ratio of the product of their extremes to the product of their means is 5 : 6. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Choose the four parts symmetrically:

Formula / Theorem Used: Four parts in AP: $(a-3d),\ (a-d),\ (a+d),\ (a+3d)$

Step 1: Finding a from Sum $= 56$ [1 Mark]

$(a-3d)+(a-d)+(a+d)+(a+3d)=56 \implies 4a=56 \implies a=14.$

Step 2: Applying the Product Ratio 5 : 6 [1.5 Marks]

$$\frac{(14-3d)(14+3d)}{(14-d)(14+d)}=\frac{5}{6} \implies \frac{196-9d^2}{196-d^2}=\frac{5}{6} \implies 6(196-9d^2)=5(196-d^2)$$

$$1{,}176-54d^2=980-5d^2 \implies 196=49d^2 \implies d^2=4 \implies d=\pm 2$$

Step 3: Listing the Four Parts [0.5 Mark]

With $d=2$: $14-6=8,\ 14-2=12,\ 14+2=16,\ 14+6=20$. (Order reverses for $d=-2$.)

Four parts: **8**, **12**, **16**, **20**

Q9. If the mth term of an AP is $\frac{1}{n}$ and its nth term is $\frac{1}{m}$ $(m \neq n)$, find its $(mn)^{\text{th}}$ term. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: $a_m = a+(m-1)d=\frac{1}{n}$ — (i) and $a_n=a+(n-1)d=\frac{1}{m}$ — (ii), with $m \neq n$.

Formula / Theorem Used: $a_k = a+(k-1)d$

Step 1: Solving for d and a [2 Marks]

Subtracting (ii) from (i): $(m-n)d=\frac{1}{n}-\frac{1}{m}=\frac{m-n}{mn} \implies d=\frac{1}{mn}$ (since $m \neq n$).

Substituting $d=\frac{1}{mn}$ into (i): $a+\frac{m-1}{mn}=\frac{1}{n} \implies a=\frac{m}{mn}-\frac{m-1}{mn}=\frac{1}{mn}$.

Step 2: Finding a_{mn} [1 Mark]

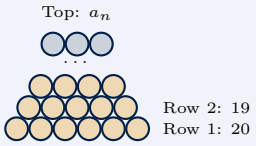
$$a_{mn}=a+(mn-1)d=\frac{1}{mn}+\frac{mn-1}{mn}=\frac{1+mn-1}{mn}=\frac{mn}{mn}=\boxed{1}$$

Q10. 200 logs are stacked in the following manner: 20 logs in the bottom row, 19 in the next row, 18 in the row next to it, and so on. In how many rows are the 200 logs placed, and how many logs are in the top row? **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Modelling Row Counts as an AP [1.5 Marks]

AP from bottom to top is $20, 19, 18, \ldots$ with $a = 20$, $d = -1$, and $S_n = 200$.



Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$ and $a_n = a + (n-1)d$

Step 2: Solving the Quadratic Equation for n [2 Marks]

$$\frac{n}{2}[2(20) + (n-1)(-1)] = 200 \implies n(41-n) = 400 \implies n^2 - 41n + 400 = 0$$

Splitting middle term ($-16 - 25 = -41$, $(-16)(-25) = 400$): $(n-16)(n-25) = 0 \implies n = 16$ or 25.

Step 3: Validating Roots and Finding Top-Row Logs [1.5 Marks]

For $n = 25$, $a_{25} = 20 - 24 = -4 < 0$ (impossible). Rejecting $n = 25$, we get $n = 16$ rows and $a_{16} = 20 - 15 = 5$.

16 rows, **5** logs in top row

Q11. The sum of four consecutive numbers in an AP is 32, and the ratio of the product of the first and last terms to the product of the two middle terms is 7 : 15. Find the numbers. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Choosing Symmetric Terms and Finding a [1.5 Marks]

Formula / Theorem Used: Four consecutive terms in AP: $(a-3d)$, $(a-d)$, $(a+d)$, $(a+3d)$

$(a-3d) + (a-d) + (a+d) + (a+3d) = 32 \implies 4a = 32 \implies a = 8$.

Step 2: Using the Product Ratio 7 : 15 to Solve for d [2.5 Marks]

$$\frac{(8-3d)(8+3d)}{(8-d)(8+d)} = \frac{7}{15} \implies \frac{64-9d^2}{64-d^2} = \frac{7}{15} \implies 15(64-9d^2) = 7(64-d^2)$$

$$960 - 135d^2 = 448 - 7d^2 \implies 960 - 448 = 135d^2 - 7d^2 \implies 512 = 128d^2 \implies d^2 = 4 \implies d = \pm 2$$

Step 3: Listing the Numbers [1 Mark]

With $d = 2$: $8 - 6 = 2$, $8 - 2 = 6$, $8 + 2 = 10$, $8 + 6 = 14$ (reversed for $d = -2$).

2, 6, 10, 14 (or **14, 10, 6, 2**)

Q12. CASE STUDY: Botanical Garden Flower Bed Planting Rows

In a botanical flower bed, there are 23 rose plants in the first row, 21 in the second row, 19 in the third row, and so on, ending with 5 rose plants in the last row. [5]

(i) Write the AP formed by the row-wise rose plants and identify its common difference d. [1]

(ii) Find the total number of rows in the flower bed. [1]

(iii) Calculate the total number of rose plants in the entire flower bed. [3]

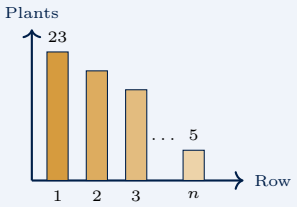
— **OR** —

How many rose plants are planted in the 5th row from the front? [3]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Extracted Scenario Data: Plant counts decrease by 2 per row from $a = 23$ down to $l = 5$.

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + l)$



Sub-part (i): Flower-Row AP and Common Difference d [1 Mark]

The AP is **23, 21, 19, …, 5** with common difference $\boxed{\mathbf{d = 21 - 23 = -2}}$.

Sub-part (ii): Total Number of Rows (n) [1 Mark]

$23 + (n-1)(-2) = 5 \implies -2(n-1) = 5 - 23 = -18 \implies n - 1 = 9 \implies \boxed{\mathbf{n = 10}\text{ rows}}$.

Sub-part (iii): Total Number of Rose Plants [3 Marks]

$$S_{10} = \frac{10}{2}(a + l) = 5(23 + 5) = 5 \times 28 = \boxed{\mathbf{140}\text{ rose plants}}$$

Sub-part (iii) [OR Option]: Rose Plants in the 5th Row [3 Marks]

$$a_5 = a + (5-1)d = 23 + 4(-2) = 23 - 8 = \boxed{\mathbf{15}\text{ rose plants}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★